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DAY3- SESSION 1

Successive Approximation Register (SAR)

Control Logic and DAC Interface for 8-Bit ADC

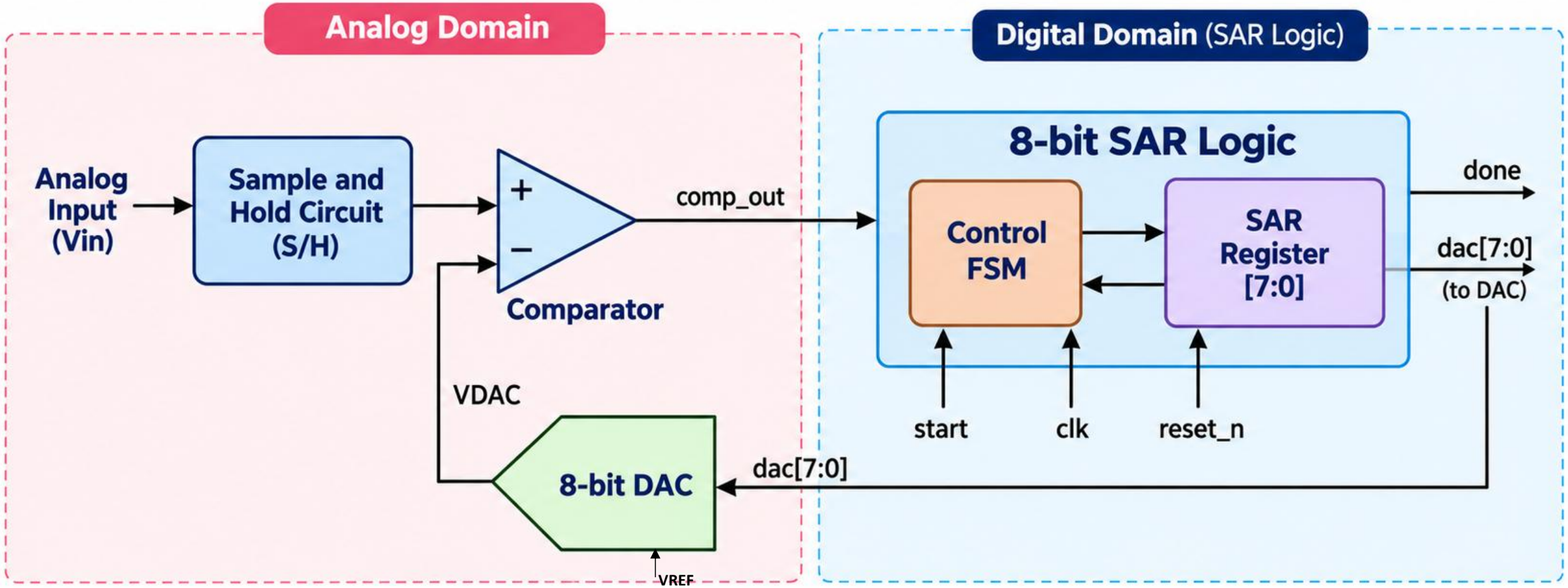
By:

SOUJANYA S

Ph.D Scholar

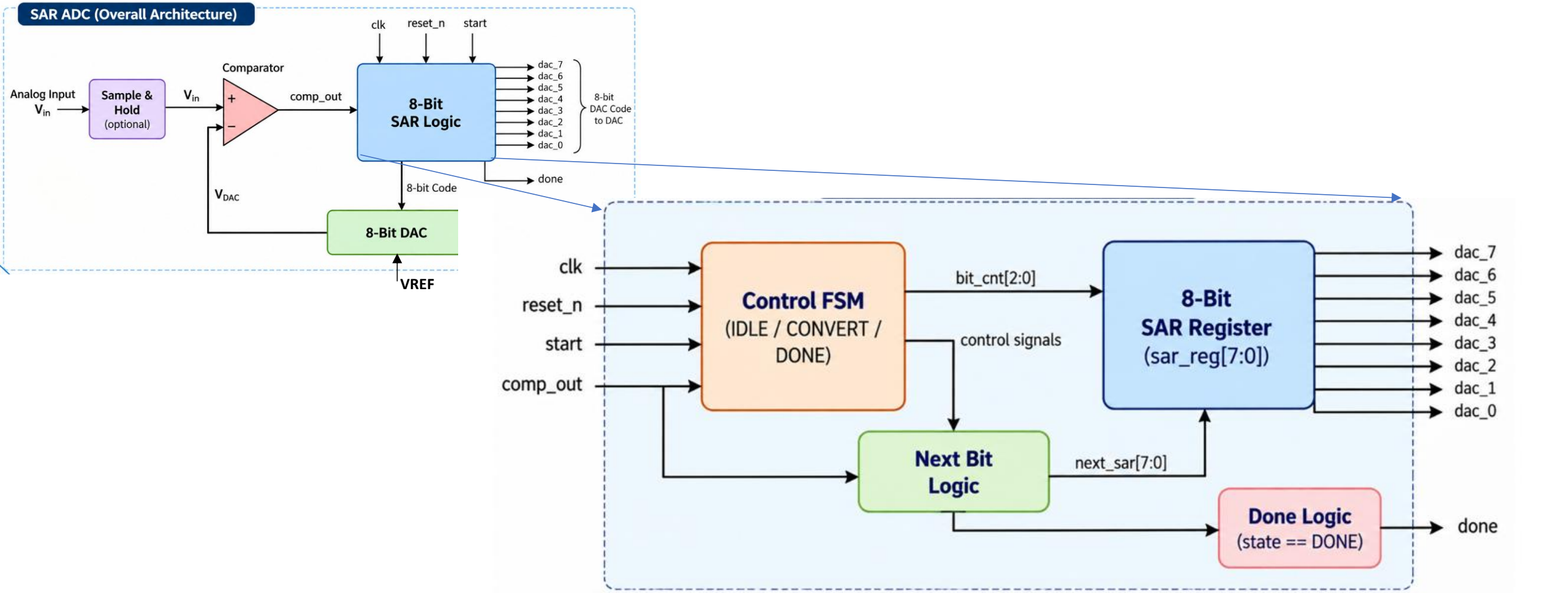
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Block Diagram – 8-Bit SAR ADC



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8 BIT SAR LOGIC BLOCK DIAGRAM



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TESTCASE

Key Parameters

$V_{IN} = 1.8\ V$

$V_{REF} = 3.3\ V$

$$V_{DAC} = V_{REF}\left[\frac{a_7}{2} + \frac{a_6}{4} + \frac{a_5}{8} + \frac{a_4}{16} + \frac{a_3}{32} + \frac{a_2}{64} + \frac{a_1}{128} + \frac{a_0}{256}\right] = \left(\frac{D}{255}\right) \times 3.3$$

$$LSB = \frac{V_{REF}}{255} = \frac{3.3}{255} = 12.94\ mV$$

→ $V_{IN} \geq V_{DAC}$ → Keep bit (1)

→ $V_{IN} < V_{DAC}$ → Clear bit (0)

D7 (MSB)
1000 0000 (128) **1.656 V**

1.8 ≥ 1.656 V (Keep 1)

D6
1100 0000 (192) **2.485 V**

0000 0000 (0) **0 V**

1.8 ≥ 2.071 V (Keep 1)

D5
1010 0000 (160) **2.071 V**

1000 0000 (128) **1.656 V**

1.8 ≥ 1.863 V (Keep 1)

D3
1000 1000 (146) **1.863 V**

1000 0000 (128) **1.656 V**

1.8 ≥ 1.811 V (Keep 1)

D2
1000 1100 (140) **1.811 V**

1000 0000 (128) **1.656 V**

1.8 ≥ 1.785 V (Keep 1)

D1
1000 1010 (138) **1.785 V**

1000 1000 (136) **1.759 V**

1.8 ≥ 1.798 V (Keep 1)

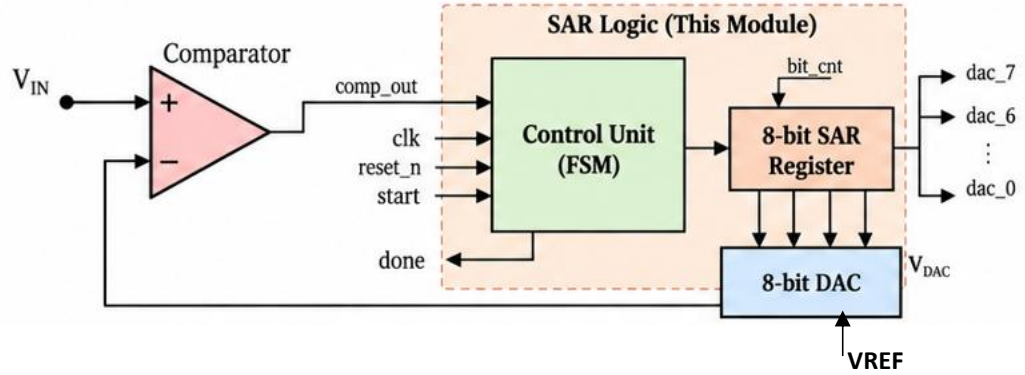
D0 (LSB)
1000 1011 (139) **1.798 V**

1000 1000 (136) **1.759 V**

Final Code

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TESTCASE ANALYSIS



DAC Model

$$V_{DAC} = \frac{DAC_CODE}{255} \times V_{REF}$$

Comparator Model

```
if (VIN ≥ VDAC)
    comp_out = 1
else
    comp_out = 0
```

For VIN = 1.8 V, VREF = 3.3 V

- Expected ADC code = (1.8 / 3.3) × 255 ≈ 139
- 139 (decimal) = 1000_1011 (binary)

Cycle (No.)	Bit Under Test (bit_cnt)	Trial Code (DAC code before decision) [7:0]	DAC Code (Decimal)	VDAC (V) (= DAC/255 × 3.3)	Compare (VIN vs VDAC)	comp_out	Action (as per code)	SAR Register (after update) [7:0]	Next Bit Set (for next trial)
0	–	0000_0000 (After reset)	0	0.0000	–	–	Reset state (IDLE)	0000_0000	–
1	7 (MSB)	1000_0000	128	1.6541	1.8 > 1.6541	1	Keep bit 7 = 1	1000_0000	Set bit 6 → 1100_0000
2	6	1100_0000	192	2.4847	1.8 < 2.4847	0	Clear bit 6 = 0	1000_0000	Set bit 5 → 1010_0000
3	5	1010_0000	160	2.0706	1.8 < 2.0706	0	Clear bit 5 = 0	1000_0000	Set bit 4 → 1001_0000
4	4	1001_0000	144	1.8624	1.8 < 1.8624	0	Clear bit 4 = 0	1000_0000	Set bit 3 → 1000_1000
5	3	1000_1000	136	1.7588	1.8 > 1.7588	1	Keep bit 3 = 1	1000_1000	Set bit 2 → 1000_1100
6	2	1000_1100	140	1.8106	1.8 < 1.8106	0	Clear bit 2 = 0	1000_1000	Set bit 1 → 1000_1010
7	1	1000_1010	138	1.7847	1.8 > 1.7847	1	Keep bit 1 = 1	1000_1010	Set bit 0 → 1000_1011
8	0 (LSB)	1000_1011	139	1.7976	1.8 > 1.7976	1	Keep bit 0 = 1	1000_1011	– (last bit)

FINAL RESULT

Final Result:

- 8-bit digital code = 1000_1101 (binary) = 139 (decimal)
- Equivalent voltage = 139 × 12.89 mV ≈ 1.794 V (close to 1.8 V)

$$LSB = \frac{V_{REF}}{255}$$

$$V_{REF} = 3.3V$$

$$LSB = \frac{3.3}{255}$$

$$LSB \approx 0.01294V$$

$$\boxed{LSB \approx 12.94mV}$$

$$D = 139$$

$$V_{DAC} = \frac{D}{255} \times V_{REF}$$

$$V_{DAC} = \frac{139}{255} \times 3.3$$

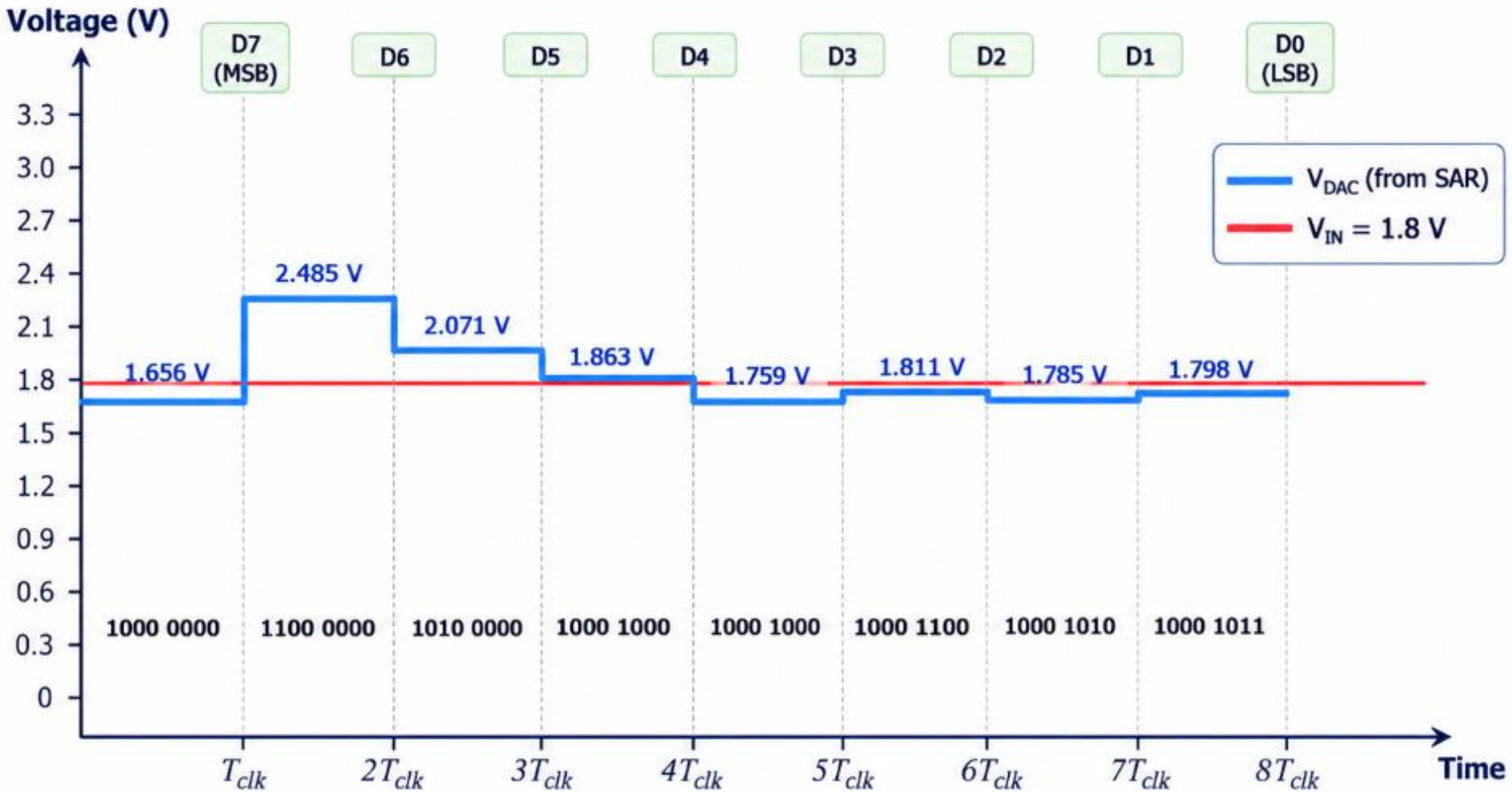
$$V_{DAC} \approx 1.7976V$$

$$\boxed{V_{DAC} \approx 1.798V}$$

$$V_{IN} = 1.8V$$

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CONVERTION TIME OF 8 BIT SAR

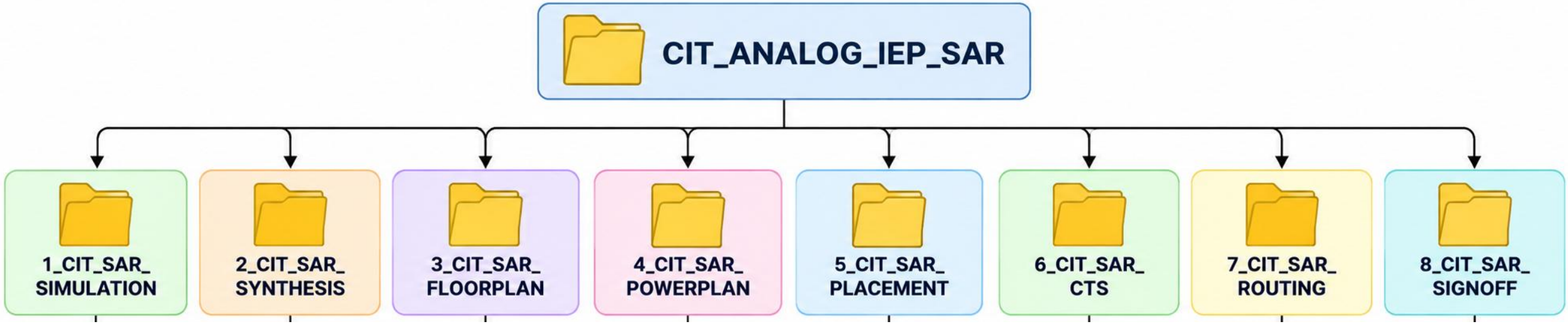


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FILE STRUCTURE



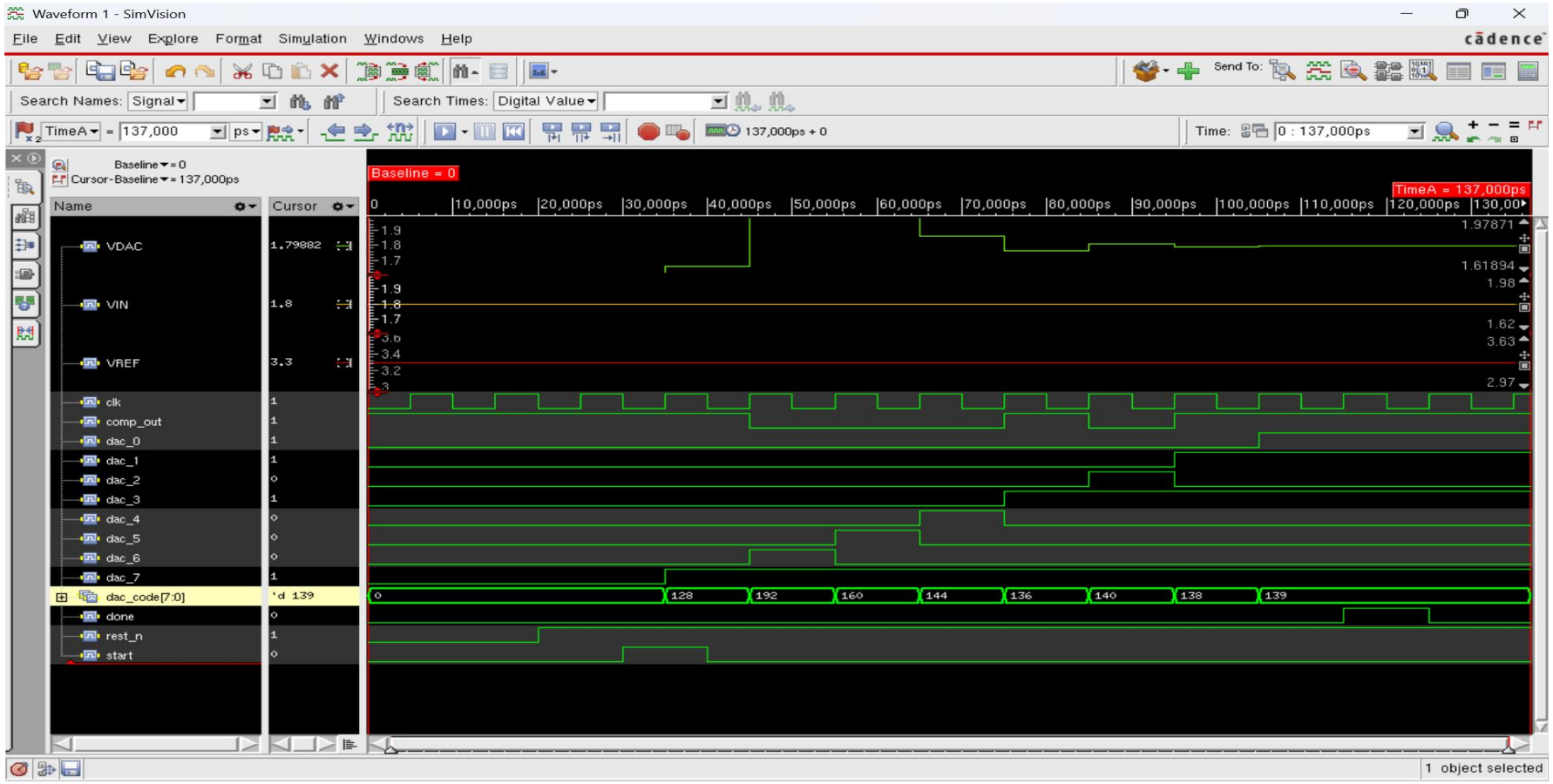


1.SIMULATION

```
[student@c2sgpr050 1_CIT_SAR_SIMULATION]$ csh
[student@c2sgpr050 1_CIT_SAR_SIMULATION]$ source /home/install/cadence/cshrc

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[student@c2sgpr050 1_CIT_SAR_SIMULATION]$ xrun sar.v sar_tb.v -access +rwc -gui
```

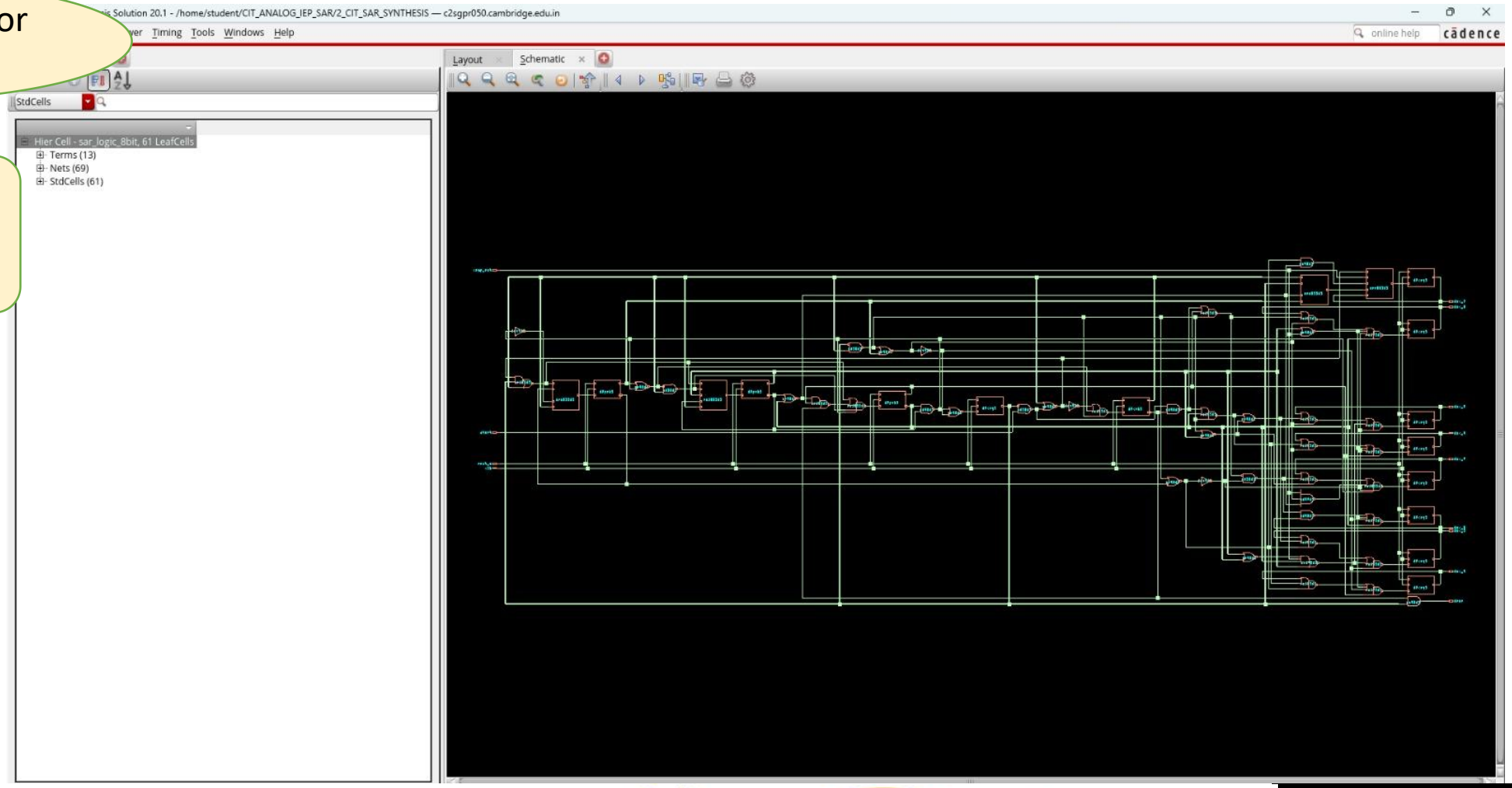


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2.SYNTHESIS

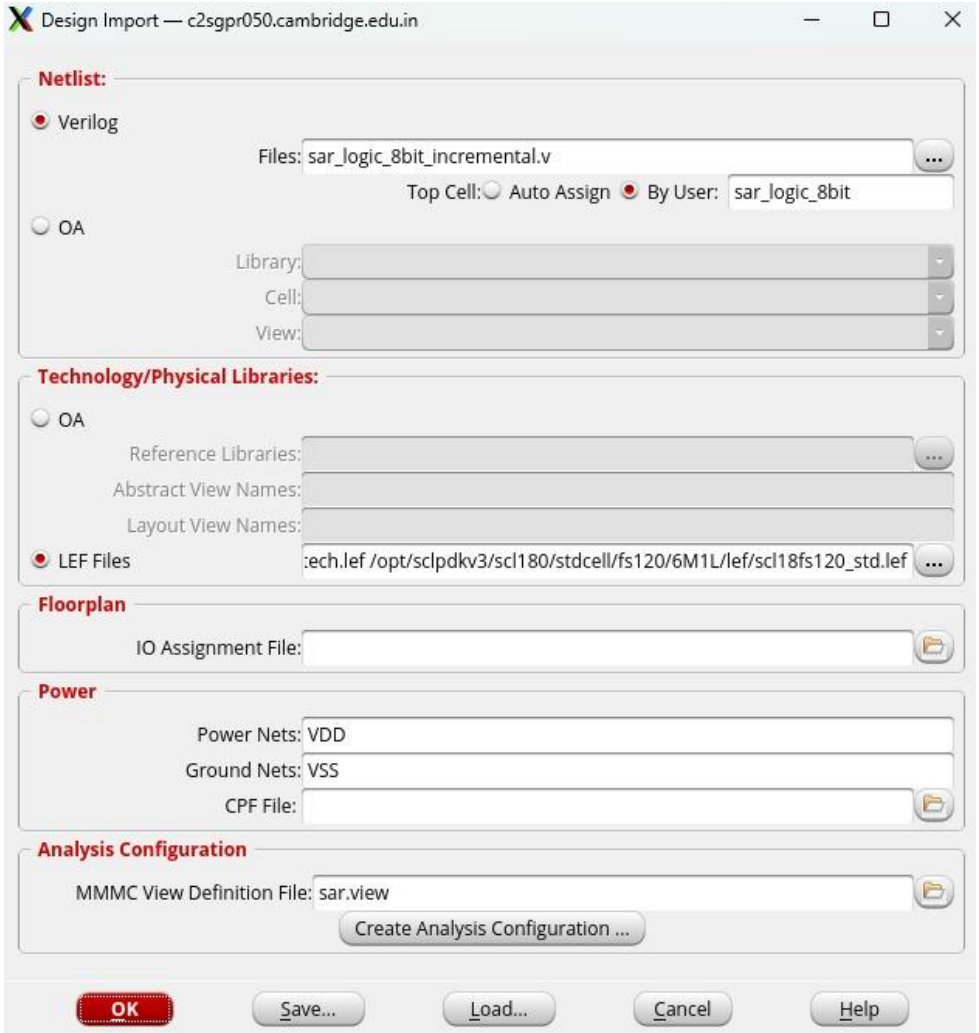
Genus tool used for synthesis

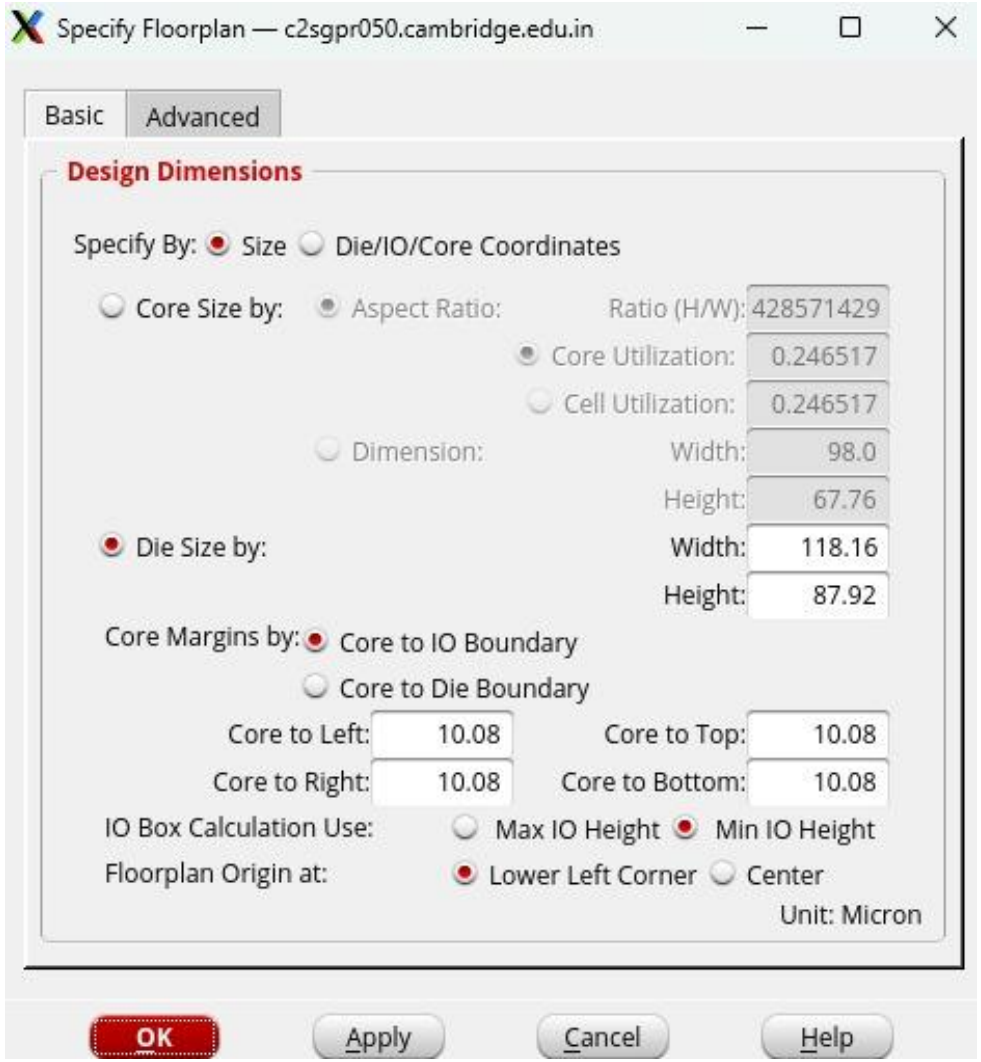
Source run.tcl



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3.FLOORPLAN





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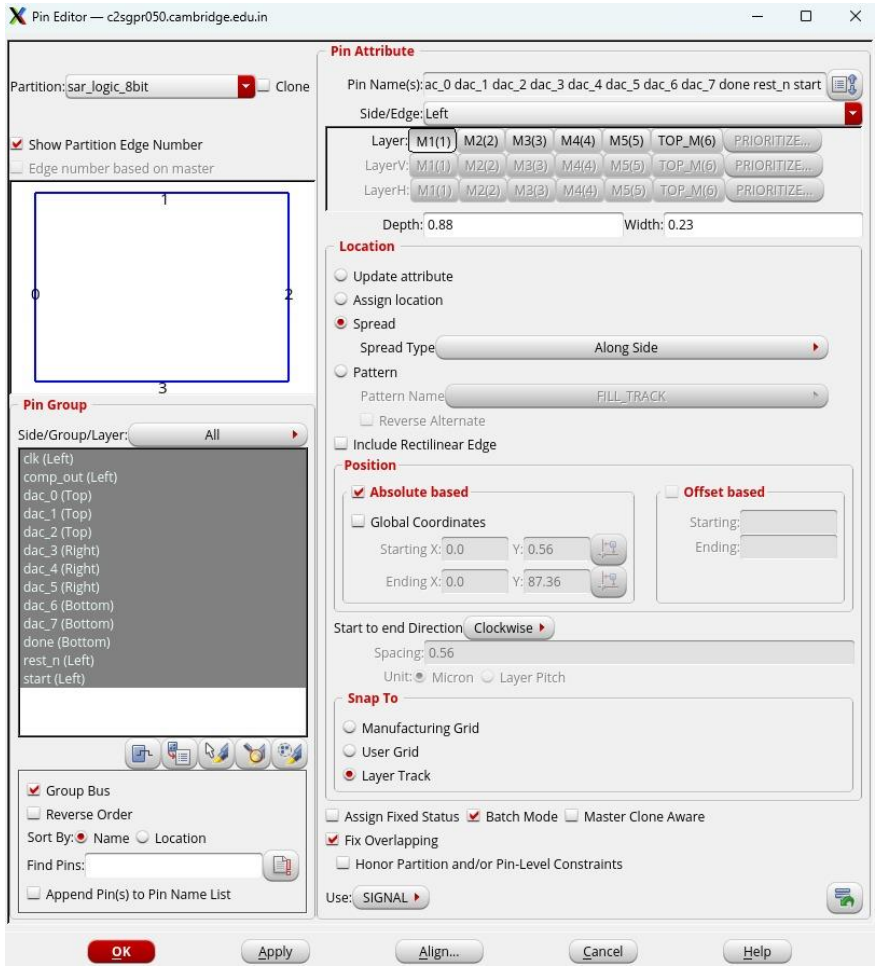
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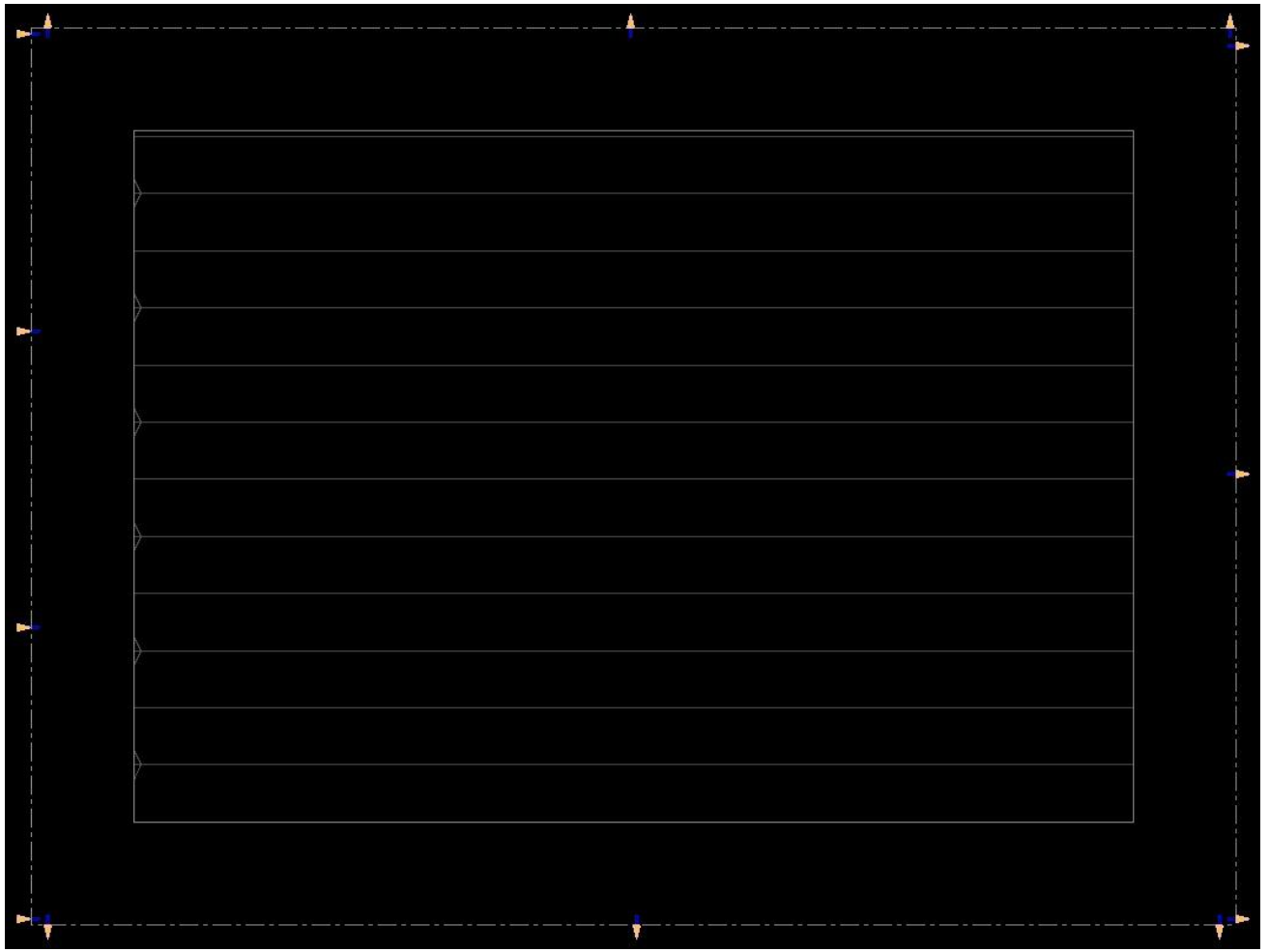
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4.POWERPLAN

```
[student@c2sgpr050 4_CIT_SAR_POWERPLAN]$ csh
[student@c2sgpr050 4_CIT_SAR_POWERPLAN]$ source /home/install/cadence/cshrc

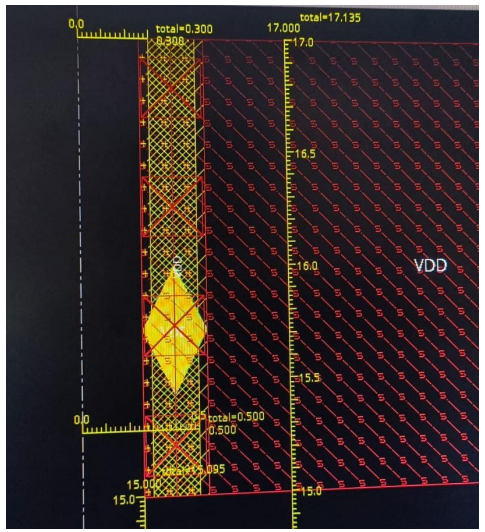
Welcome to Cadence Tools Suite

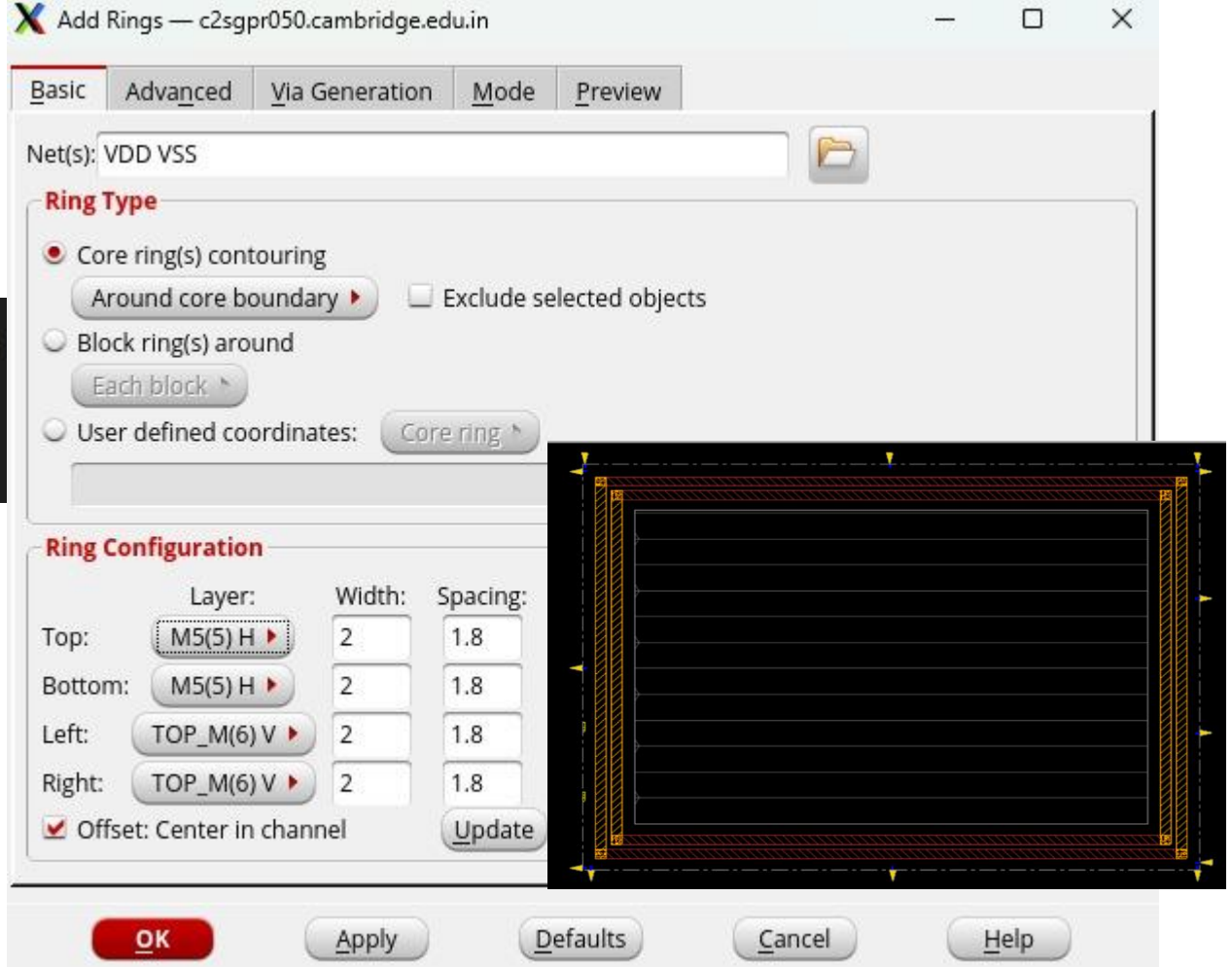
[student@c2sgpr050 4_CIT_SAR_POWERPLAN]$ innovus
```

```
globalNetConnect VDD -type pgpin -pin VDD -all
globalNetConnect VSS -type pgpin -pin VSS -all
globalNetConnect VDD -type tiehi
globalNetConnect VSS -type tielo


# VDD Pin (Y = 15.0 to 17.0)
createPGPin VDD -net VDD -geom M4 0.30 15.0 0.5 17.0

# VSS Pin (Y = 30.0 to 32.0)
createPGPin VSS -net VSS -geom M4 0.30 30.0 0.5 32.0
```



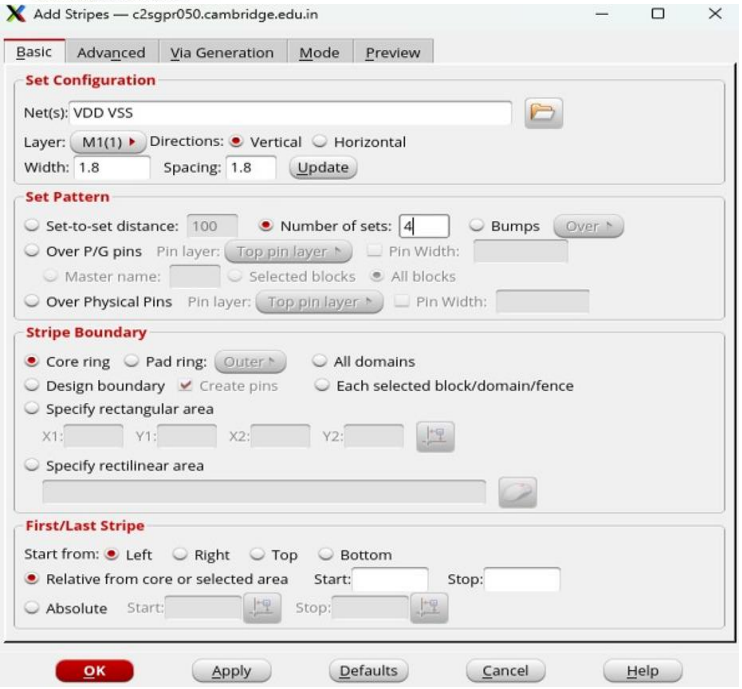


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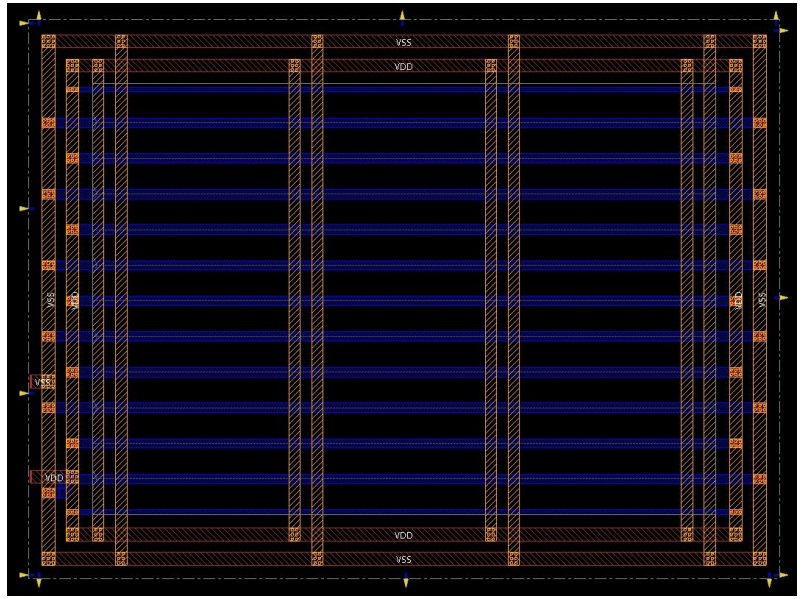
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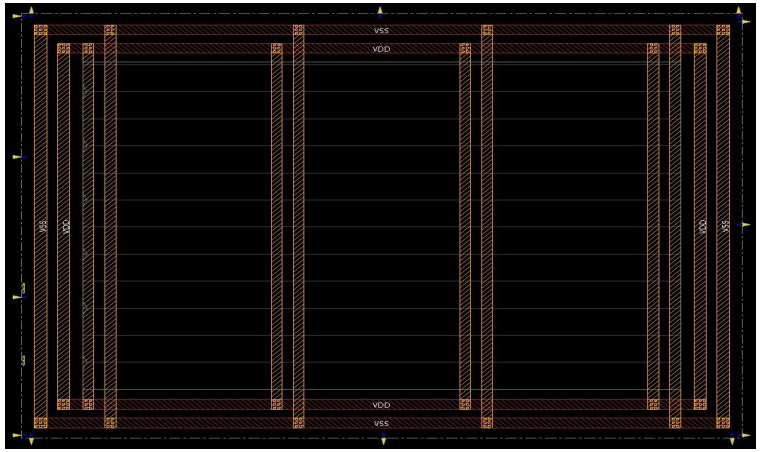
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4.POWERPLAN









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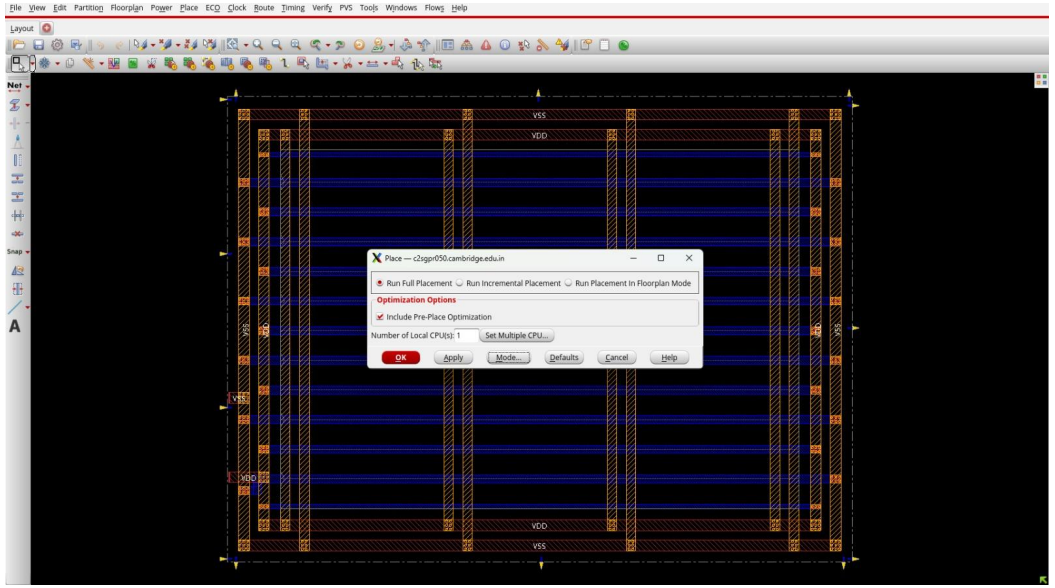
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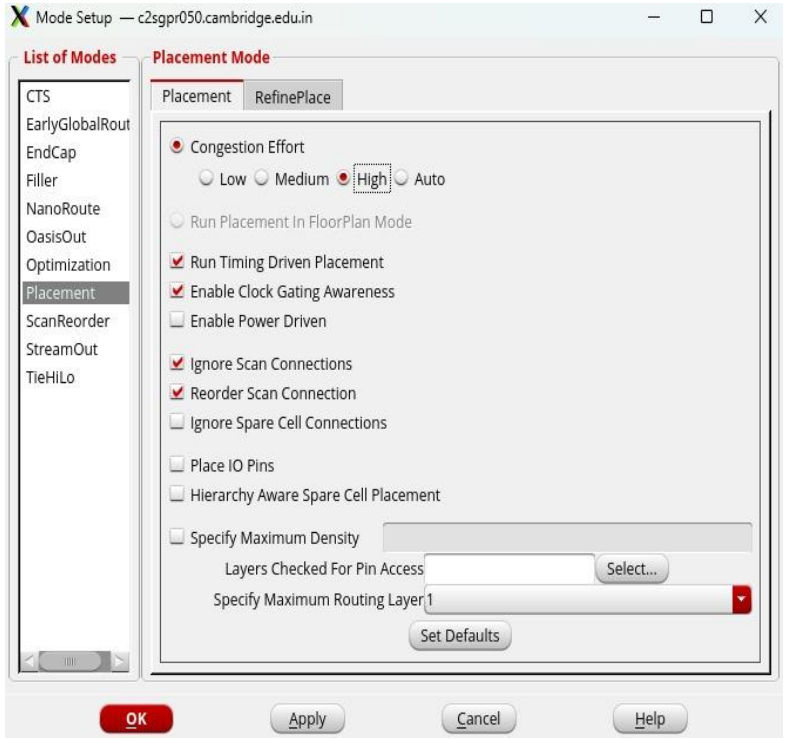
5.PLACEMENT

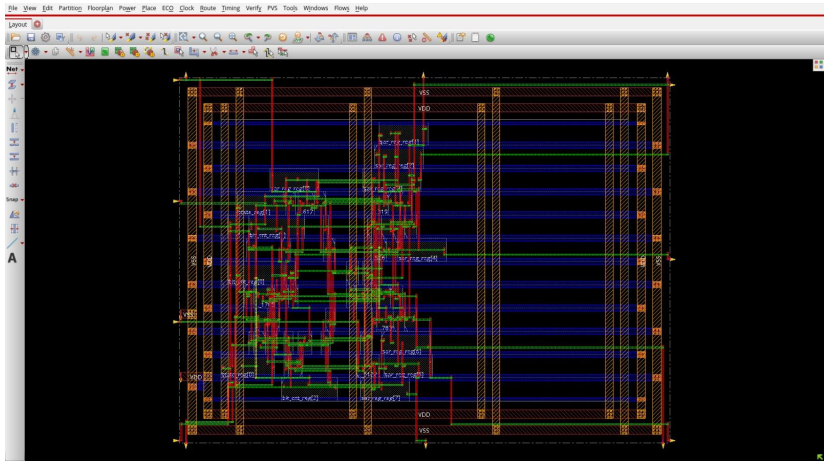
```
[student@c2sgpr050 5_CIT_SAR_PLACEMENT]$ csh
[student@c2sgpr050 5_CIT_SAR_PLACEMENT]$ source /home/install/cadence/cshrc

Welcome to Cadence Tools Suite

[student@c2sgpr050 5_CIT_SAR_PLACEMENT]$ innovus
```







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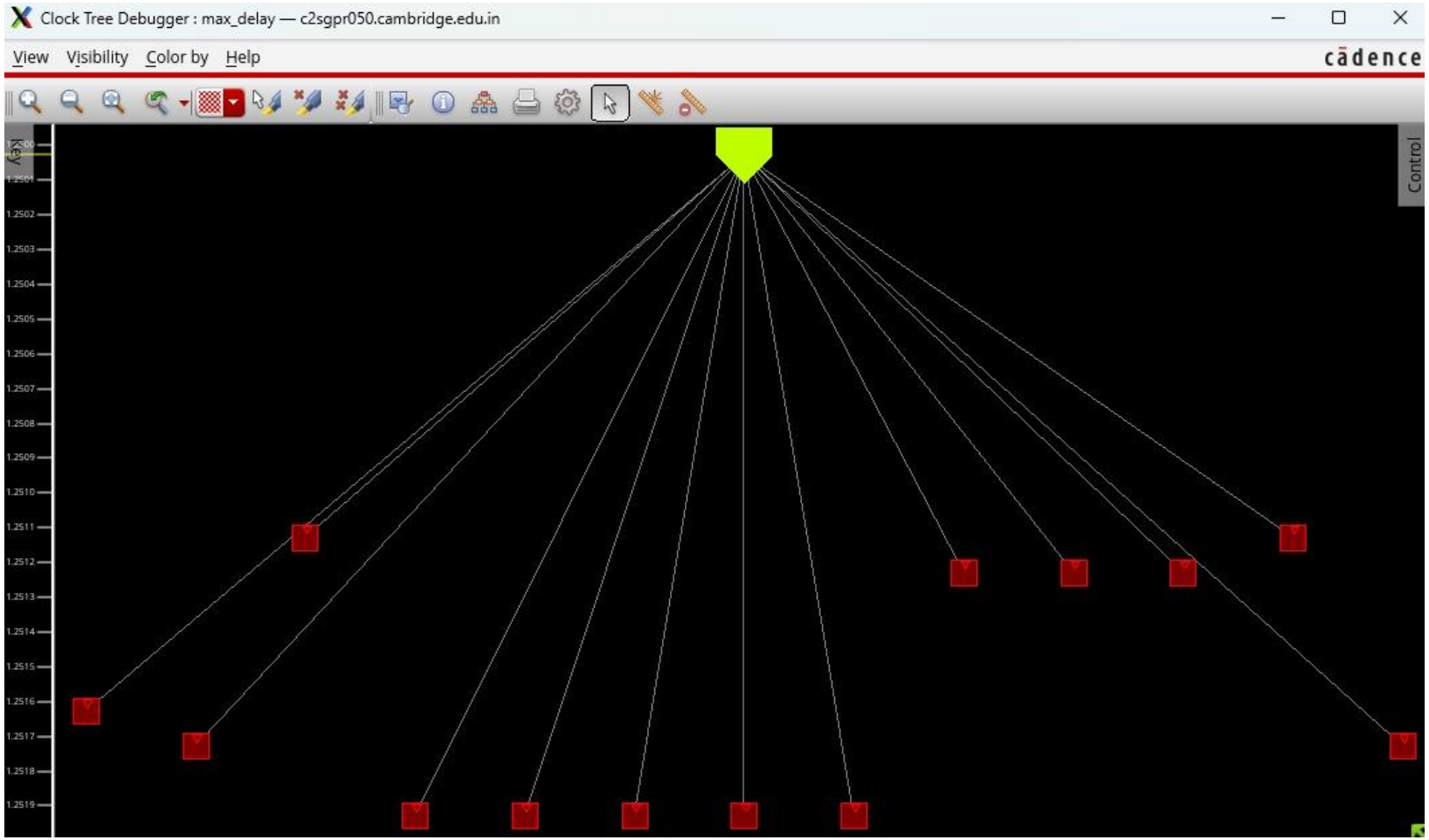
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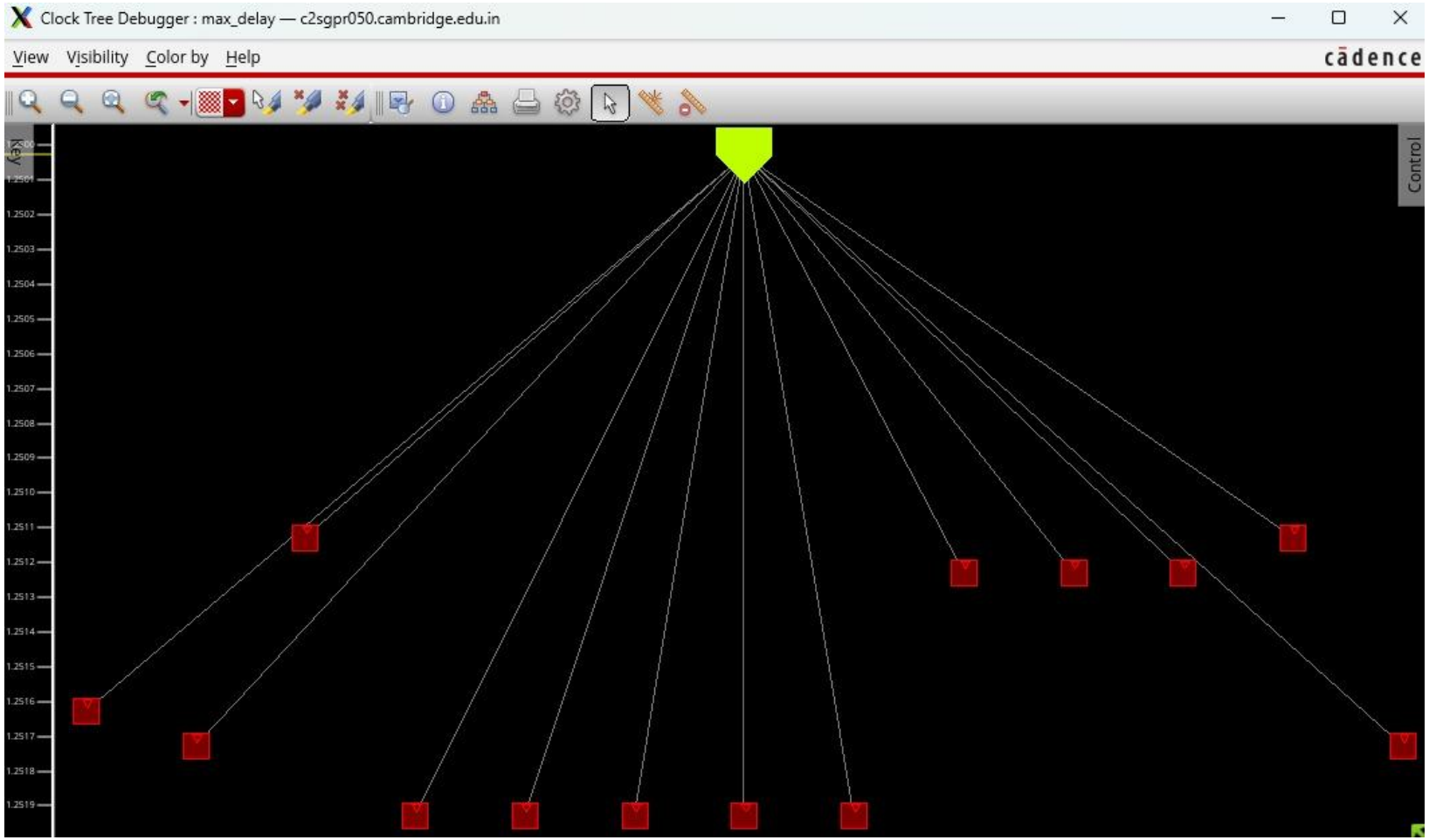
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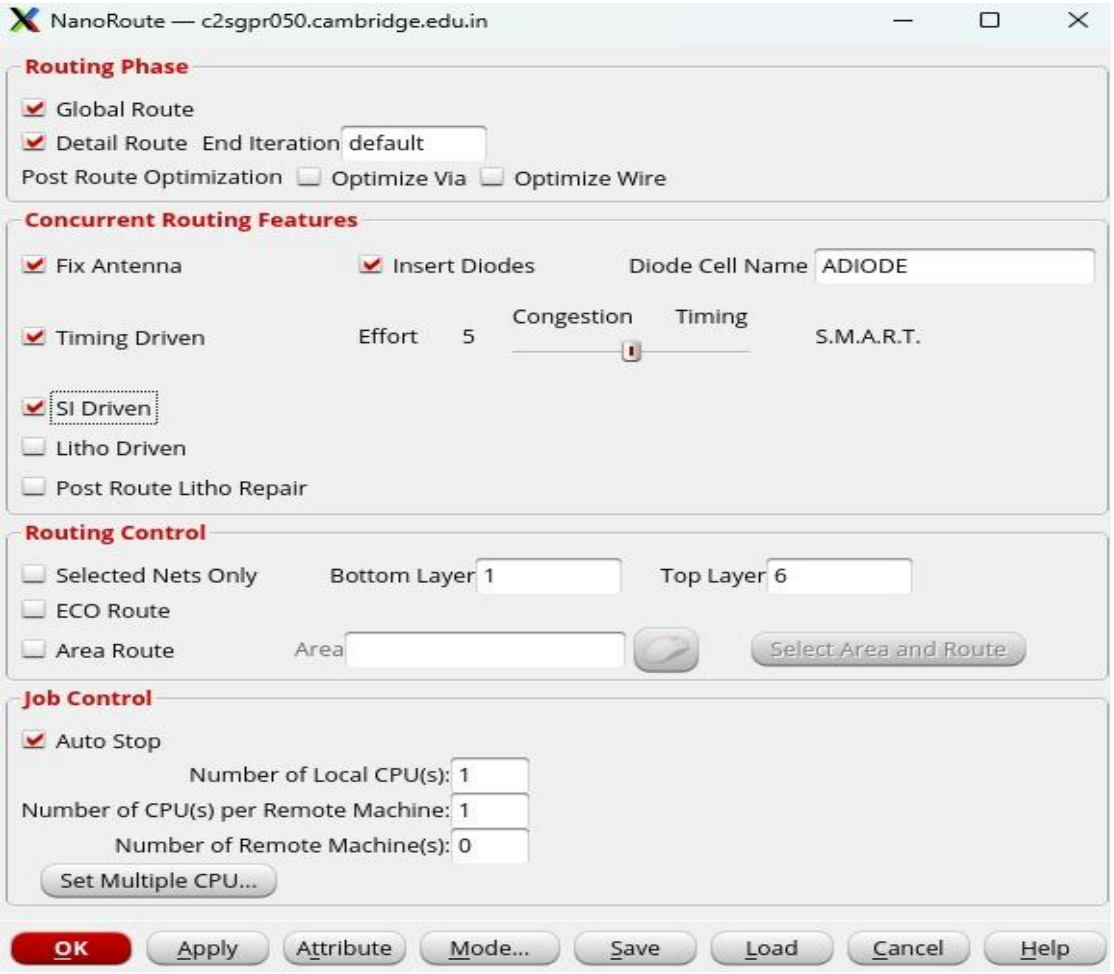
6. CLOCK TREE SYNTHESIS

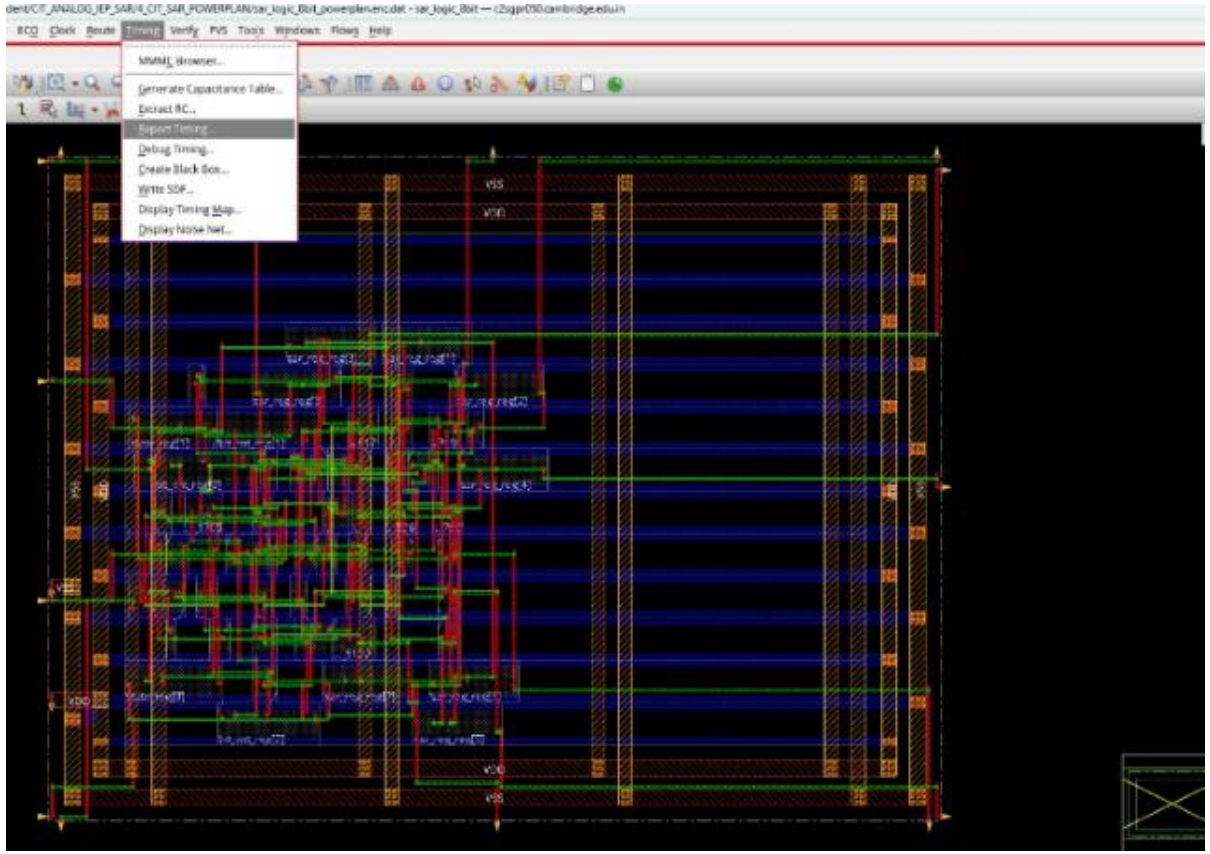


6. CLOCK TREE SYNTHESIS



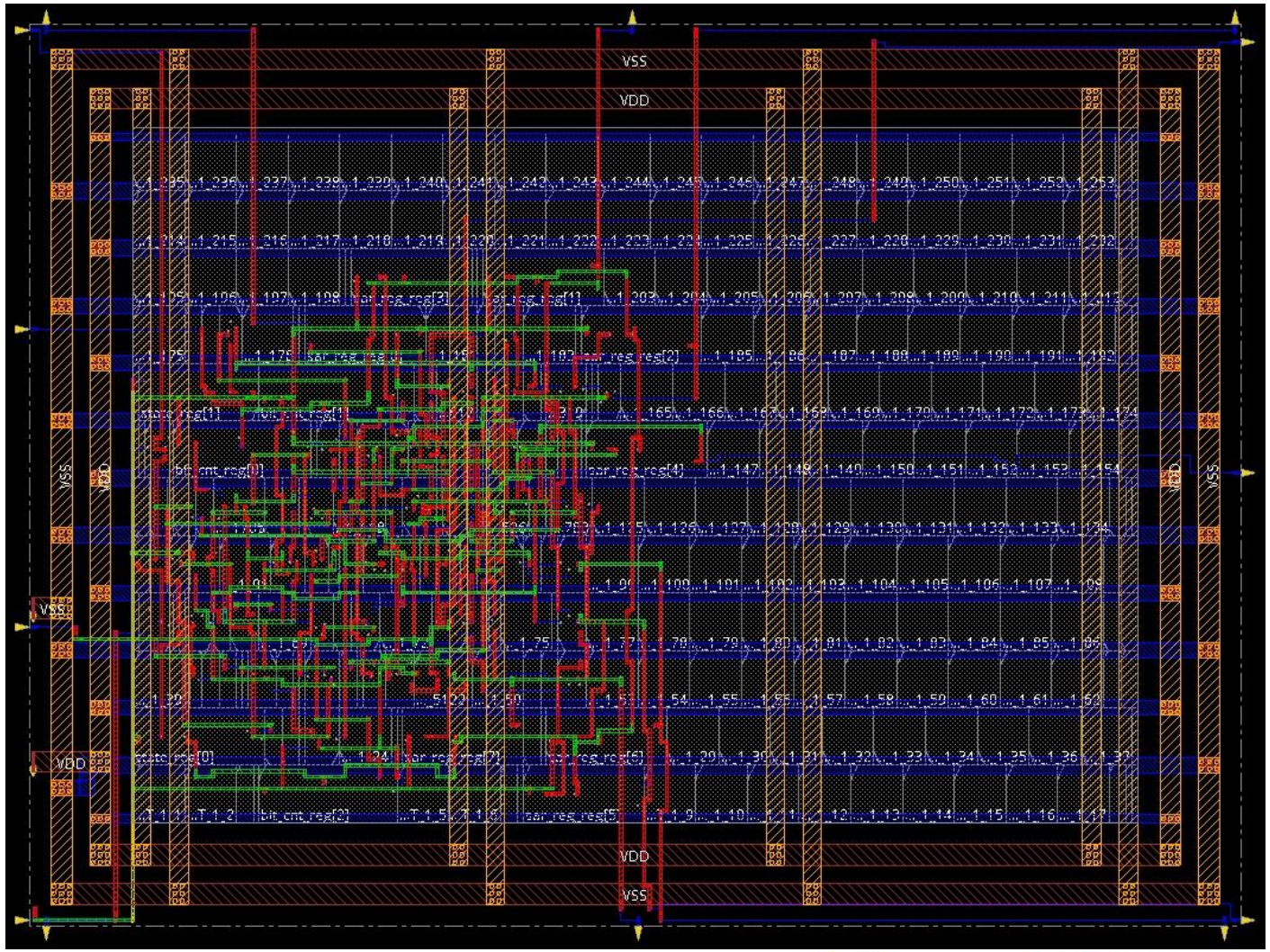
7. ROUTING





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7. ROUTING



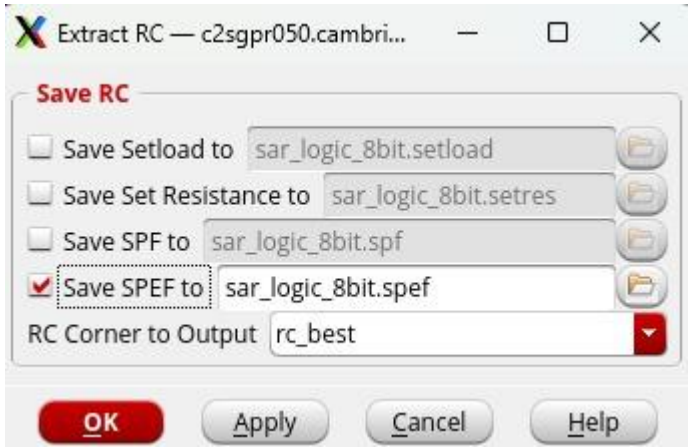
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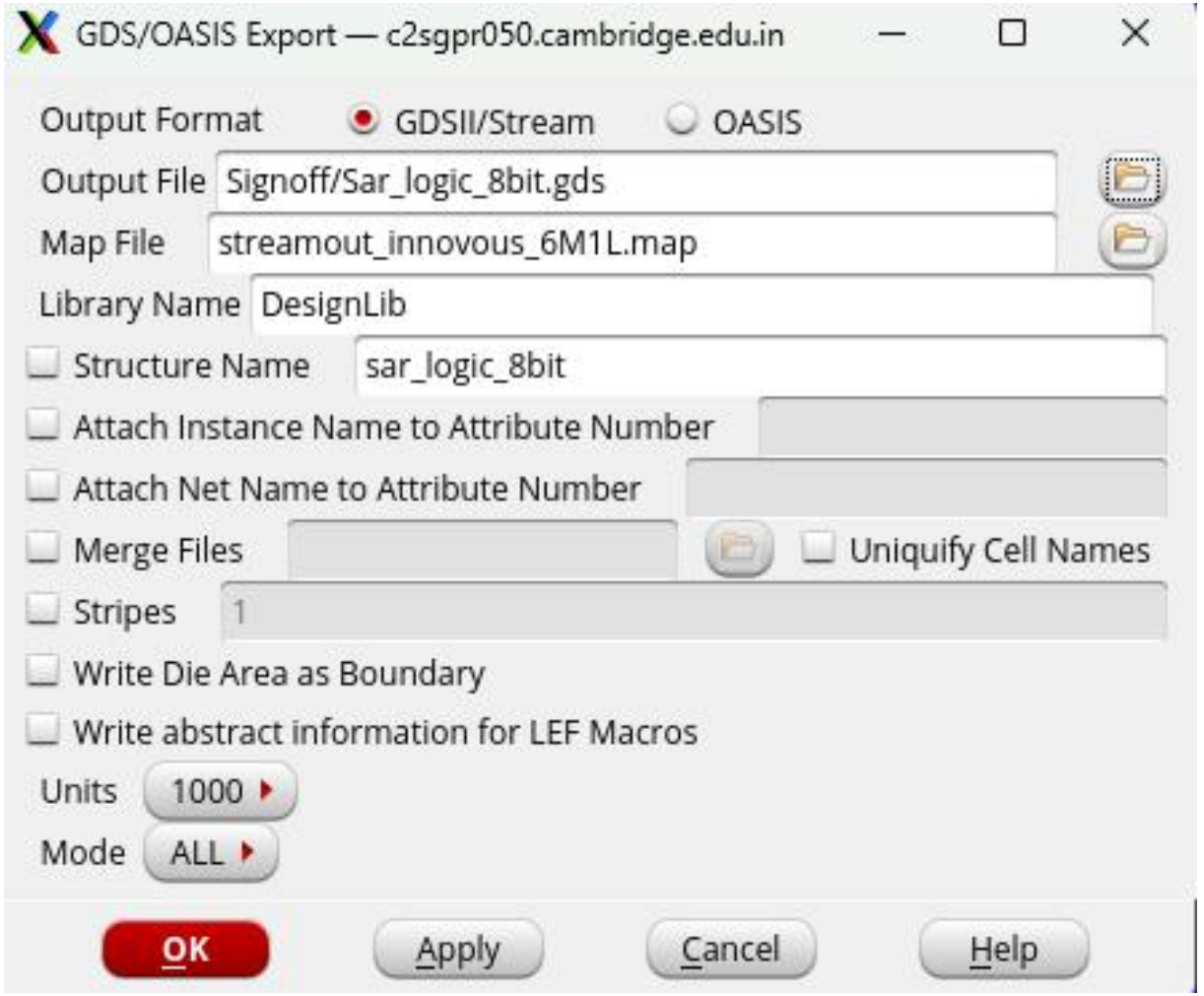
8. SIGNOFF

```
[student@c2sgpr050 8_CIT_SAR_SIGNOFF]$ csh
[student@c2sgpr050 8_CIT_SAR_SIGNOFF]$ source /home/install/cadence/cshrc

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[student@c2sgpr050 8_CIT_SAR_SIGNOFF]$ innovus
```






```
innovus 1> verify_drc
 *** Starting Verify DRC (MEM: 1282.9) ***

  VERIFY DRC ..... Starting Verification
  VERIFY DRC ..... Initializing
  VERIFY DRC ..... Deleting Existing Violations
  VERIFY DRC ..... Creating Sub-Areas
  VERIFY DRC ..... Using new threading
  VERIFY DRC ..... Sub-Area: {0.000 0.000 118.160 87.920} 1 of 1
  VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.

  Verification Complete : 0 Viols.

 *** End Verify DRC (CPU: 0:00:00.0  ELAPSED TIME: 0.00  MEM: 4.0M) ***

1
```


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Thank You!

Any Questions?

